

Multiple Choice (10 marks)

1. In reference to a segment of DNA, which of the following molecules contains the fewest number of nucleotides?
 - (a) the tRNA transcript from the original DNA
 - (b) a single strand of the original DNA segment after a substitution mutation
 - (c) the final processed mRNA made from the original DNA
 - (d) the primary RNA transcript (after splicing) from the original DNA
 - (e) a single strand of the original DNA segment after a duplication mutation
2. Which of the following statements about mitochondria and chloroplasts is generally true?
 - (a) Plants, animals, and fungi all have both chloroplasts and mitochondria.
 - (b) Plants and fungi have chloroplasts but no mitochondria; animals have only mitochondria.
 - (c) Animals have both chloroplasts and mitochondria; plants and fungi have only chloroplasts.
 - (d) Plants and animals have chloroplasts but no mitochondria; fungi have only mitochondria.
 - (e) Plants and animals have mitochondria but no chloroplasts; fungi have only chloroplasts.
3. During the period when life is believed to have begun, the atmosphere on primitive Earth contained abundant amounts of all the following gases EXCEPT
 - (a) hydrogen
 - (b) methane
 - (c) nitrogen
 - (d) sulfur dioxide
 - (e) neon
4. Why does milk "when kept in a refrigerator?"
 - (a) Enzymatic reactions with oxygen
 - (b) Lactose crystallization
 - (c) Loss of refrigeration power over time
 - (d) mesophiles

- (e) Natural separation of milk components without bacterial influence
- 5. What are the typical periods in the unfolding of a behavioral act?
 - (a) Cue, Desire, Fulfillment
 - (b) Arousal, Behavior, Consequence
 - (c) Stimulus, Habituation, Extinction
 - (d) Appetitive behavior, Consummatory act, Quiescence
 - (e) Perception, Decision, Movement

True / False (10 marks)

Answer True or False

- 6. True or False: Starting insulin in type 2 diabetes: continue oral hypoglycemic agents?
- 7. True or False: Production of chemokines by perivascular adipose tissue: a role in the pathogenesis of atherosclerosis?
- 8. True or False: The English antibiotic awareness campaigns: did they change the public's knowledge of and attitudes to antibiotic use?
- 9. True or False: Biomolecular identification of allergenic pollen: a new perspective for aerobiological monitoring?
- 10. True or False: Should temperature be monitored during kidney allograft preservation?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the mechanisms by which certain animals maintain a constant body temperature despite fluctuations in environmental temperatures, focusing on physiological adaptations and ecological implications.
- 12. Discuss how stabilizing selection influences reproductive success in Swiss starlings, particularly in relation to clutch size and fitness outcomes.

End of Paper